Physical Chemistry Cume
(based on literature: Science, 2023, 382, 99-103)
10/20/23

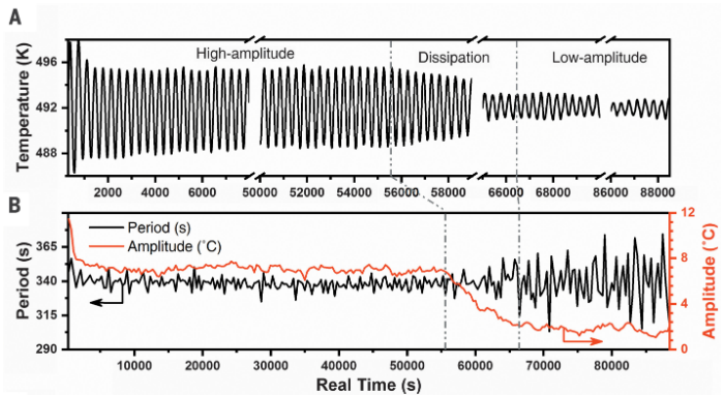
1. (25 pts) Answer the following short questions:

   a) What is "syngas'?
   b) How to produce "syngas" from coal gasification?
   c) Write the water gas shift reaction associated with "syngas".
   d) What is the Fischer-Tropsch (FT) reaction?
   e) Give two examples of active catalysts for the FT reaction.

2. (25 pts) The figure to the right shows the oscillatory behavior in the FT reaction using Ar as the carrier gas.

   a) Describe the observations.

   b) Explain why no oscillatory behavior was observed after replacing Ar with He.

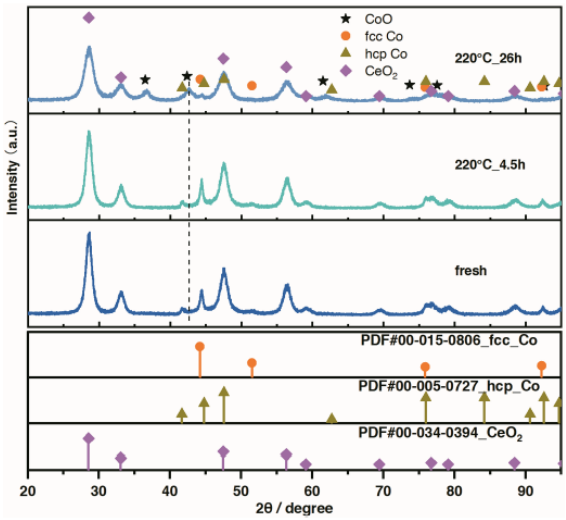


3. (25 pts) Write the FT reaction elementary step and the rate law of the following two pathways:

   a) the CO insertion into O-H bonds of the surface hydroxy.
   b) the CO insertion into O-R bonds of the surface alkoxy.

4. (25 pts) The figure to the right shows the XRD patterns of a solid sample before and after different treatments.

   a) Describe the structural/compositional changes due to different treatments.

   b) What does "fcc" and "hcp" stand for?

   c) What is the packing density of "fcc Co" and "hcp Co", respectively? Show your work.